

# Course Project Guide

Work in teams to complete a data mining project, write a report, create a poster, and present your work at the Poster Session. Regardless of the topic chosen, your project should follow a complete data mining workflow, including data acquisition (collecting data or obtaining an existing dataset), data preprocessing, task definition, experimental design and implementation, comparison of different methods, analysis of results and conclusions, and visualization. The final poster should clearly communicate the problem, approach, key findings, and limitations.

## 1. Project Reference Examples

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The four projects below are illustrative examples, not prescribed assignments. Their application domains, data sources, problem definitions, and suggested methods are provided for reference only. Your group may adapt these examples or propose your own project in any domain, provided that it addresses at least one of the four data mining task types: classification, clustering, anomaly detection, or pattern mining.

### 1.1 Clustering: Social Media Community and Topic Evolution

Select a public topic or event and obtain relevant data from a major social-media platform such as Weibo, Xiaohongshu, Zhihu, Douyin, or Bilibili. You may use an existing public dataset or collect publicly accessible data yourself using appropriate tools. Based on the posts, users, interactions, and other available information, formulate a meaningful clustering problem. Possible problems include discovering different discussion topics, user groups, or communities. Design suitable representations, build and compare different clustering approaches, investigate how the discovered groups differ under different representations or stages of the event, and derive meaningful conclusions about how the selected online discussion is organized and evolves over time.

### 1.2 Classification: Robust Human Mobility Classification

Obtain a public human-mobility dataset, such as the Microsoft GeoLife GPS Trajectories dataset, and formulate a meaningful classification problem based on the available trajectory or mobility information. Possible problems include recognizing transportation modes such as walking, cycling, driving, or public transportation. Process the raw data, design suitable features or learned representations, and build and compare different classification models. Investigate the models under different realistic conditions, such as unseen users, reduced sampling frequency, noisy or incomplete data, or different feature representations, and derive conclusions about what factors affect classification accuracy, robustness, and generalization.

### 1.3 Anomaly Detection: Spatio-Temporal Urban Sensor Anomaly Detection

Obtain a public multi-location sensor dataset, such as the UCI Beijing Multi-Site Air Quality dataset, and formulate a meaningful anomaly-detection problem. Possible problems include detecting measurements that deviate from their historical patterns, identifying readings inconsistent with nearby monitoring stations, or discovering unusual environmental events. Define a reasonable evaluation setting, build and compare different anomaly-detection approaches using temporal, spatial, or combined information, and investigate their performance under different types of abnormal behavior or experimental conditions. Analyze representative successes and failures and derive conclusions about when different sources of contextual information are useful for reliable anomaly detection.

### 1.4 Pattern Mining: Urban Mobility Pattern Discovery

Obtain a public urban-mobility dataset, such as Citi Bike trip histories or NYC Taxi trip records, and formulate a meaningful pattern-mining problem from the available spatial-temporal data. Possible problems include discovering frequent origin-destination flows, recurring commuting behaviors, or common mobility sequences. Transform the raw records into suitable transactions, sequences, regions, or mobility events, apply and compare appropriate pattern-mining methods or settings, and examine how the discovered patterns differ across contexts such as weekdays and weekends, rush and non-rush hours, seasons, or urban regions. Analyze representative patterns and derive meaningful conclusions about recurring human-mobility behavior and how it changes under different conditions.

## 2. Project Report

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Report length: 6–10 pages of main text, including figures and tables; references, the disclosure section, and appendices are excluded.

Submit a detailed report (Report.pdf) introducing your implementation algorithms and summarizing the experimental setup, observations, and analyses for each experiment. Please follow the structure below to organize your project report:

- **Title & Group Members:** Project title, group number, and team member names/IDs.
- **1. Introduction:** Real-world context and motivation, analytical objectives, and main contributions.
- **2. Problem Formulation:** Formal task definition, objectives, inputs, outputs, and key terms.
- **3. Methodology:** Data preprocessing, the main methods, system architecture/pipeline, key design choices, and implementation details.
- **4. Experiments:** Dataset source/schema and size, evaluation protocol, baseline methods, metrics, parameter settings, and comparative result analysis; include data splits where applicable.
- **5. Conclusion and Limitations:** Key discoveries, practical implications, limitations, and future improvements.

6. Disclosures: Identify external collaborators and describe how AIGC was used in the project.

References: List the sources cited in the report. Do not include fabricated references.

- Appendix: Supplementary results or technical details, if applicable.

## 3. Project Poster

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Prepare Poster.pdf for the Poster Session, an interactive presentation session where students and instructors walk around to view different posters, while team members introduce and pitch their project to visiting audiences. Because visitors will browse through multiple projects, your poster must serve as a clear, standalone visual story. It should concisely convey the entire project narrative from problem motivation to methodology, results, and practical impact. It provides key visual evidence that audiences can grasp quickly, while serving as a visual anchor during your team's live walkthrough and defense.

Poster formatting requirements and sample posters will be released later.

## 4. Data Sources

The following five public sources provide starting points for finding data. These are references, not prescribed datasets.

Weibo-COV — Weibo text and timestamps for topic clustering. [GitHub repository](#)

GeoLife GPS Trajectories — mobility data for the classification example. [Microsoft dataset page](#)

Beijing Multi-Site Air Quality — sensor data for the anomaly-detection example. [UCI dataset page](#)

NYC TLC Trip Record Data — taxi trips for the pattern-mining example. [TLC data page](#)

Citi Bike System Data — bike trips for the pattern-mining example. [Citi Bike data page](#)

## 5. Submission

Once your team is confirmed, any one member should submit the proposal, including the project title, a brief introduction, and all team members' names, by Oct 15 (Thu 23:59), the same deadline as team registration. The following week, through Oct 22, is reserved for proposal review.

For the final submission, use Group\_XX\_Project.zip, replacing XX with your Canvas group identifier. Include Report.pdf, the code, a README, and data or download instructions. Poster.pdf is submitted separately before the session and is not required in the final ZIP.

Provide the full dataset or download instructions. If special circumstances prevent you from providing the full original data, include sample data and explain the reason in the README. The README should document dependencies, data preparation, and commands to run the code. For a sample-data exception, explain which results the sample run can demonstrate and which full-data results it cannot reproduce.

Submissions up to one week late incur a deduction of 20% of the earned mark (final mark = earned mark  $\times$  0.8).

## 6. Academic Integrity

Do not submit directly AI-generated work or fabricated references. In a separate Disclosures section of Report.pdf, identify external collaborators and describe how AIGC was used.